

420 Final Project - MC Variance Reduction

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October 2, 2026

What is Monte Carlo Integration?

- Use random sampling to approximate difficult integrals
- Replace deterministic computation with simulation
- Example:

$$\theta = \int_S f(x)g(x) dx = \mathbb{E}_f[g(X)], \quad X \sim f$$

- Can be approximated using:

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n g(X_i), \quad X_i \sim f$$

- Useful when analytical solutions are difficult

Motivation

- Since we are using samples, this leads to variance in our estimates
- The variance introduced leads to unstable estimates
- Intuitive solution:
 - Increase sample size
- Although:
 - This is rather expensive computationally
- So how do we reduce variance without simply increasing n ?

Variance Reduction Techniques

- Antithetic sampling reduces variance through negative correlation
- We now consider two additional methods:
- Stratified Sampling
 - Improves coverage of the domain
- Importance Sampling
 - Focuses sampling in regions that contribute most

Importance Sampling: Motivation

- Standard Monte Carlo often samples uniformly
- Inefficient when integrand is more complicated
- Many samples contribute little to estimate
- This creates high variance
- Goal:
- Sample from a distribution that matches the shape of the integrand

Importance Sampling Estimator

We want to estimate:

$$\theta = \int_S g(x)f(x) dx$$

Rewrite:

$$\theta = \int_S g(x) \frac{f(x)}{\phi(x)} \phi(x) dx = \mathbb{E}_{\phi} \left[\frac{g(X)f(X)}{\phi(X)} \right]$$

Estimator:

$$\hat{\theta}^I = \frac{1}{n} \sum_{i=1}^n \frac{g(X_i)f(X_i)}{\phi(X_i)}, \quad X_i \sim \phi$$

Key Idea and Variance

Variance:

$$\text{Var}(\hat{\theta}') = \frac{1}{n} \left(\int \frac{(g(x)f(x))^2}{\phi(x)} dx - \theta^2 \right)$$

Key Insight:

- Variance depends on $\frac{g(x)f(x)}{\phi(x)}$
- If ratio is nearly constant then we achieve low variance

Goal:

- Choose $\phi(x)$ to resemble $g(x)f(x)$

Importance Sampling Algorithm

Goal: Estimate $\theta = \int_S g(x)f(x)dx$

① Sample $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \phi$

② Compute:

$$h(X_i) = \frac{g(X_i)f(X_i)}{\phi(X_i)}, \quad i \in (1, 2, \dots, n)$$

③ Return:

$$\hat{\theta}^I = \frac{1}{n} \sum_{i=1}^n h(X_i)$$

Stratified Sampling: Motivation

- Standard Monte Carlo samples uniformly
- Relying on random variance can still lead to clustering
- Some regions may receive fewer samples than others
- **Uneven coverage creates high variance**
- **Idea:**
- Split the support into smaller sections that encourage more uniform coverage

Stratified Sampling Estimator

We want to estimate:

$$\theta = \int_a^b g(x) dx$$

Rewrite:

$$\theta = \sum_{j=1}^k \int_{I_j} g(x) dx$$

where

$$I_j = \left[a + \frac{j-1}{k}(b-a), a + \frac{j}{k}(b-a) \right]$$

Estimator:

$$\hat{\theta} = \sum_{j=1}^k \frac{b-a}{k} \cdot \frac{1}{r} \sum_{i=1}^r g(X_{ji}), \quad X_{ji} \sim \text{Uniform}(I_j)$$

Key Idea and Variance

Variance:

$$\text{Var}(\hat{\theta}) = \frac{(b-a)^2}{k^2} \sum_{j=1}^k \frac{\sigma_j^2}{r}$$

where $\sigma_j^2 = \text{Var}(g(X) \mid X \in I_j)$

Key Insight:

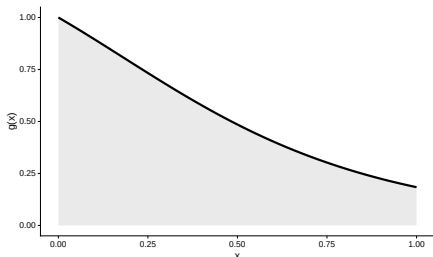
- Variance depends on variability within each stratum
- If $g(x)$ is nearly constant within each stratum, variance is low

Goal:

- Splice the support so that $g(x)$ is more uniform within each stratum

Example I : Stratified Sampling

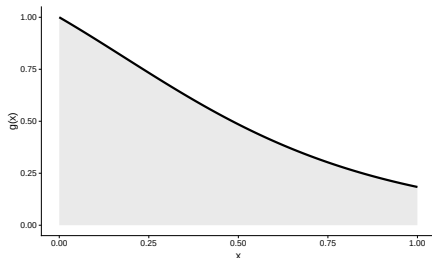
$$\theta = \int_0^1 \frac{e^{-x}}{1+x^2} dx$$



- Pseudocode for estimating theta
 - 1 Input n, m strata, optional seed
 - 2 Divide interval [0,1] into m equal length intervals
 - 3 For each stratum $j = 1, 2, \dots, m$, Generate $U_j = U_{j1}, U_{j2}, \dots, U_{jk} \stackrel{\text{iid}}{\sim} \text{Uniform}(0,1)$, $k = \frac{n}{m}$
 - 4 Transform to the jth stratum: $X_{ji} = \frac{j-1}{m} + \frac{U_{ji}}{m}$
 - 5 Evaluate $\hat{\theta}_j = \frac{1}{k} \sum_{i=1}^k \frac{e^{-X_{ji}}}{1+X_{ji}^2}$
 - 6 Return combination of the strata estimates $\hat{\theta}_S = \frac{1}{m} \sum_{j=1}^m \hat{\theta}_j$

Example I : Importance Sampling

$$\theta = \int_0^1 \frac{e^{-x}}{1+x^2} dx$$



- Pseudocode for estimating theta using an exponential proposal distribution $\phi(x) \sim \text{EXP}(1)$
 - 1 Inputs: n, seed
 - 2 Sample $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(1)$.
 - 3 Return

$$\hat{\theta}^I = \frac{1}{n} \sum_{i=1}^n \frac{1}{1+X_i^2} \mathbf{1}_{(0,1)}(X_i).$$

Example I: Results

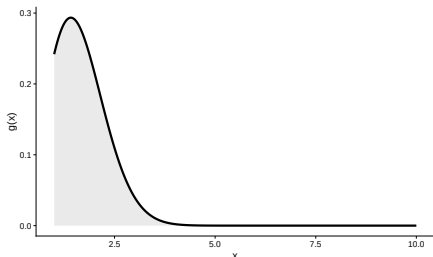
The results for Example I are found in the following table.

Method	Mean Estimate	Variance	Relative Efficiency
Standard MC	0.5247	5.868e-05	1.00
Stratified MC	0.5248	6.227e-07	94.23
Importance MC	0.5248	1.672e-04	0.35
Antithetic MC	0.5248	2.138e-06	27.45

Table: Example 6.13 - Mean, Variance, and Relative Efficiency

Example II: Stratified Sampling

$$\theta = \int_1^{1000} \frac{x^2}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

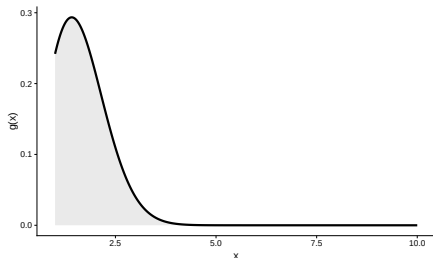


- Pseudocode for estimating theta

- 1 Input n, m strata, optional seed
- 2 Divide interval [1,1000] into m equal length intervals
- 3 For each stratum $j = 1, 2, \dots, m$, Generate $U_j = U_{j1}, U_{j2}, \dots, U_{jk} \stackrel{\text{iid}}{\sim} \text{Uniform}(0,1)$, $k = \frac{n}{m}$
- 4 Transform to the j th stratum: $X_{ji} = \frac{j-1}{m} + \frac{U_{ji}}{m}$
- 5 Evaluate $\hat{\theta}_j = \frac{1}{k} \sum_{i=1}^k \frac{X_{ji}^2}{\sqrt{2\pi}} e^{-X_{ji}^2/2}$
- 6 Return combination of the strata estimates $\hat{\theta}_S = \frac{1}{m} \sum_{j=1}^m \hat{\theta}_j$

Example II: Importance Sampling

$$\theta = \int_1^{1000} \frac{x^2}{\sqrt{2\pi}} e^{-x^2/2} dx.$$



- Pseudocode for estimating theta using an exponential proposal distribution $\phi(x) \sim \text{EXP}(1)$
 - 1 Inputs: n , seed.
 - 2 Sample $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(1)$.
 - 3 Compute and Return

$$\hat{\theta}' = \frac{1}{n} \sum_{i=1}^n \frac{X_i^2}{\sqrt{2\pi}} \exp\left(-\frac{X_i^2}{2} + X_i\right) \mathbf{1}_{(1,1000)}(X_i)$$

Example II: Results

The results for Example II are found in the following table.

Method	Mean Estimate	Variance	Relative Efficiency
Standard MC	0.4044	9.070e-02	1.00
Stratified MC	0.4007	8.590e-02	1.05
Importance MC	0.4007	3.307e-04	274.25
Antithetic MC	0.4092	8.823e-02	1.03

Table: Example II - Mean, Variance, and Relative Efficiency

- **Example 1:**

- Stratified sampling reduces variance from 5.868×10^{-5} to 6.227×10^{-7}
- Importance sampling does not reduce variance in this setting

- **Example 2:**

- Importance sampling reduces variance from 9.070×10^{-2} to 3.307×10^{-4}
- Stratified sampling provides only a slight reduction

- **Key Point:**

- Stratified and importance sampling reduce variance by improving how samples are distributed

Conclusion

- Standard Monte Carlo methods are limited by high variance
- Stratified and importance sampling improve estimator efficiency
 - Achieve lower variance without increasing sample size
- Results show:
 - Stratified sampling is effective when uniform coverage is needed
 - Importance sampling is effective when key regions dominate the integral
- **Takeaway:**
 - Variance reduction methods allow more accurate Monte Carlo estimates with fewer samples

Limitations and Future Work

- Performance depends on parameter choices
 - Number of strata
 - Choice of importance distribution
- Methods are problem-dependent
 - Not all techniques perform well in every setting
- Future Work:
 - Explore better selection of proposal distributions
 - Apply methods to more complex or higher-dimensional problems

Acknowledgements Q/A

- Rizzo, M. (2019). *Statistical Computing with R*
- Course notes and materials from STAT 420
- Dr. Bell for guidance and instruction

Thank you. Questions?